






Optimal Robust Pulses for Atomic Fountain Interferometry

Michael Goerz[1], P. Kunz[1], M. Kasevich[2], V. Malinovsky[1]

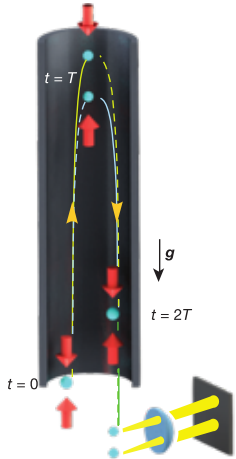
[1]U.S. Army Research Lab, [2]Stanford University

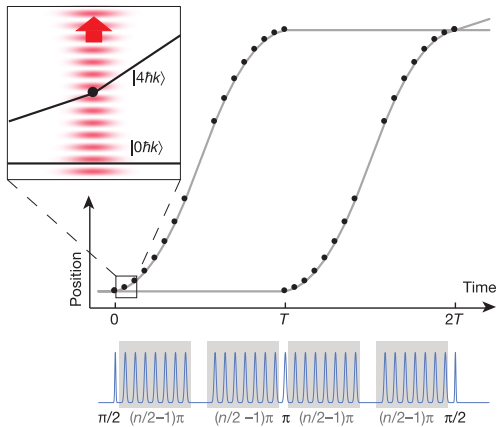
DAMOP Meeting

June 2020

UNCLASSIFIED

10 m atomic fountain at Stanford (Kasevich lab): ultracold ^{87}Rb atomic cloud

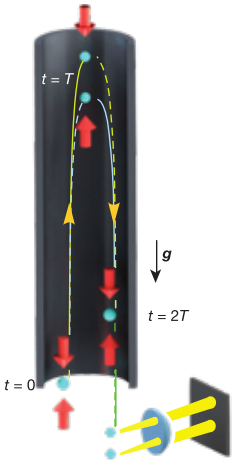




$$\Delta\phi = -2k_{\max}gT^2$$

Kovachi et al. *Nature* **528**, 530 (2015)

10 m atomic fountain at Stanford (Kasevich lab): ultracold ^{87}Rb atomic cloud

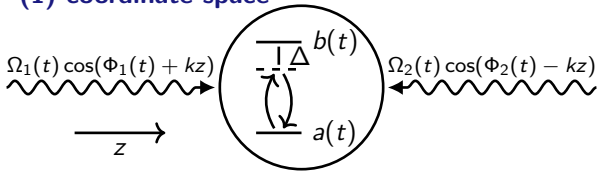


Applications:

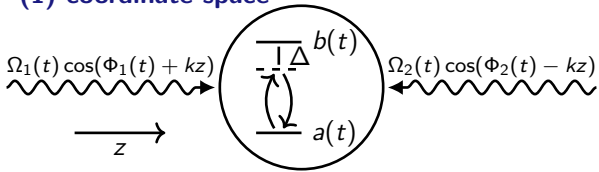
- inertial navigation
- gravitational sensing
- test of equivalence principle

$$\Delta\phi = -2k_{\mathrm{max}}gT^2$$

(1) coordinate space



(1) coordinate space

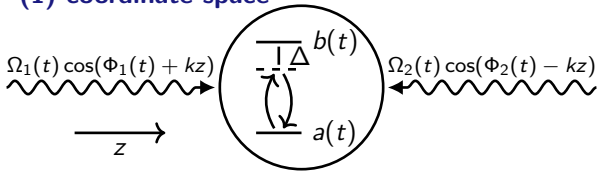


$\Downarrow$ (2) adiabatic elimination of $b(t)$ — RWA

effective amplitude $\Omega(t) = \Omega_1(t)\Omega_2(t)/4\Delta$

effective phase $\phi(t) = \Phi_1(t) - \Phi_2(t)$

(1) coordinate space

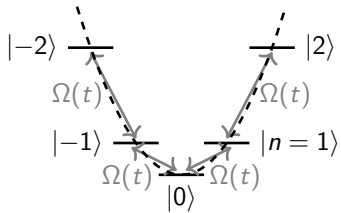


⇓ (2) adiabatic elimination of $b(t)$ — RWA

$$\text{effective amplitude } \Omega(t) = \Omega_1(t)\Omega_2(t)/4\Delta$$

$$\text{effective phase } \phi(t) = \Phi_1(t) - \Phi_2(t)$$

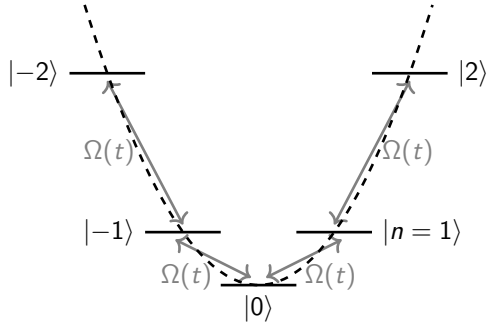
⇓ (3) discretization in momentum space (initial momentum p_0)



$$E_n(t) \approx n^2\hbar\omega_k + 2n\beta\hbar\omega_k + n\hbar\dot{\phi}(t))$$

$$\omega_k = 2\hbar k^2/m$$

$$\beta = p_0/2\hbar k$$



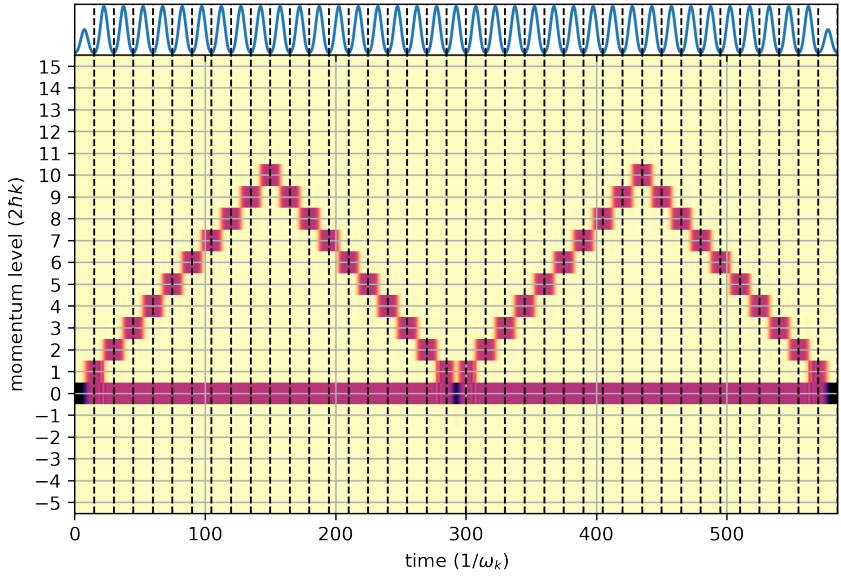
$$E_n(t) \approx n^2\hbar\omega_k + 2n\beta\hbar\omega_k + n\hbar\dot{\phi}(t))$$

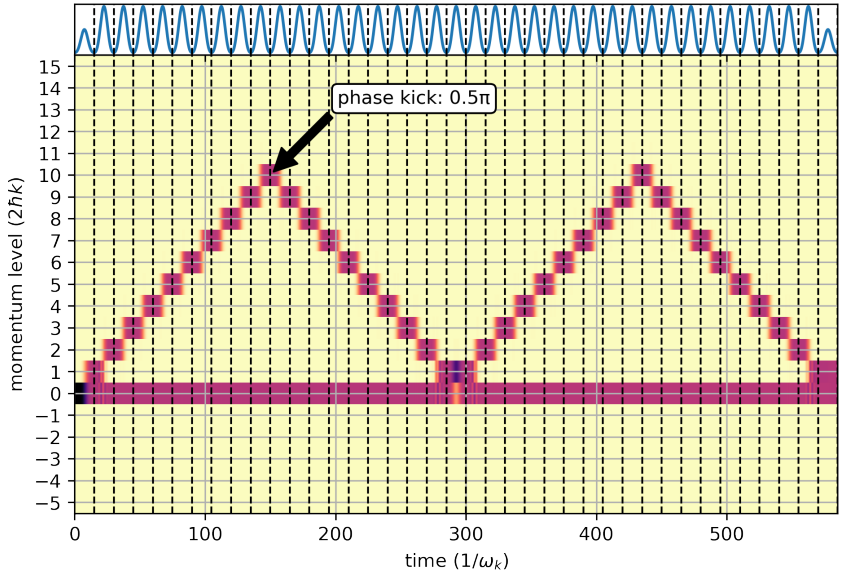
$$\omega_k = 2\hbar k^2/m$$
$$\beta = p_0/2\hbar k$$

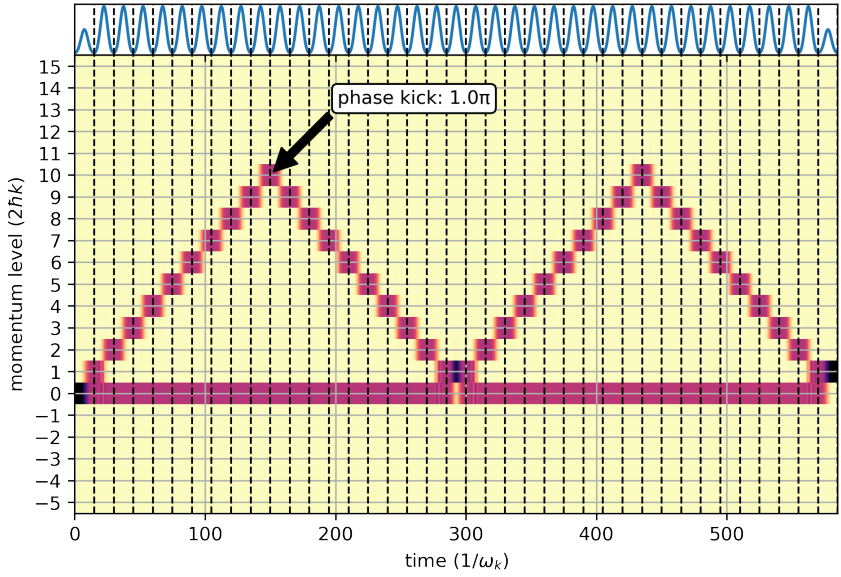
Train of Rabi pulses:

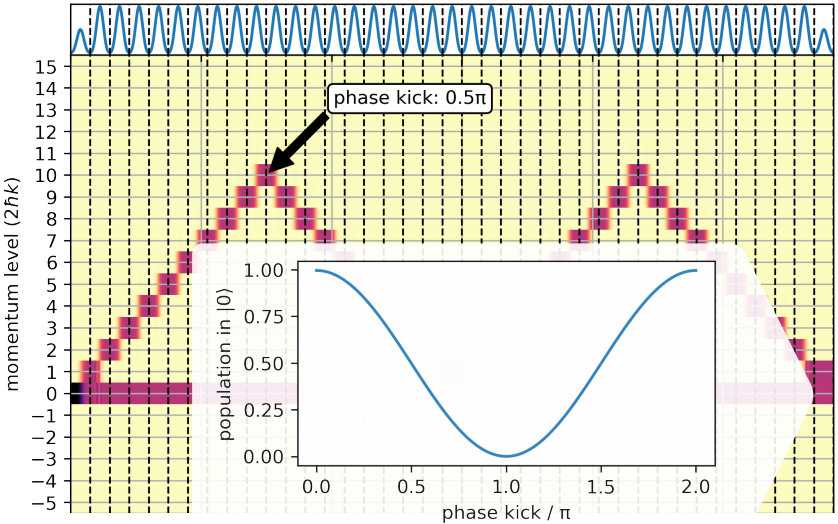
Choose laser frequency such that levels n_0 and $n_0 + 1$ are shifted into resonance.

$$\phi(t) = \omega_L t; \qquad \omega_L = -\omega_k(2n_0 + 1)$$


momentum level ($2\hbar k$)
time ($1/\omega_k$)
15
14
13
12
11
10
9
8
7
6
5
4
3
2
1
0
−1
−2
−3
−4
−5
0
100
200
300
400
500


phase kick: 0.5π
momentum level ($2\hbar k$)
time ($1/\omega_k$)
15
14
13
12
11
10
9
8
7
6
5
4
3
2
1
0
−1
−2
−3
−4
−5
0
100
200
300
400
500


phase kick: 1.0π
momentum level ($2\hbar k$)
time ($1/\omega_k$)
15
14
13
12
11
10
9
8
7
6
5
4
3
2
1
0
−1
−2
−3
−4
−5
0
100
200
300
400
500


phase kick: 0.5π
momentum level (2ℏk)
15
14
13
12
11
10
9
8
7
6
5
4
3
2
1
0
−1
−2
−3
−4
−5
population in |0⟩
1.00
0.75
0.50
0.25
0.00
0.0
0.5
1.0
1.5
2.0
phase kick / π

Robustness?

$$\hat{\mathrm{H}} = \sum_n \left(E_n^{(0)}(t) + 2n\beta\hbar\omega_k\right)|n\rangle\langle n| - \Omega(t)\left(|n\rangle\langle n+1| + \mathrm{H.c.}\right)$$

Robustness?

$$\hat{\mathrm{H}} = \sum_n \left(E_n^{(0)}(t) + 2n\beta\hbar\omega_k \right) |n\rangle\langle n| - \mu S(t) \left(|n\rangle\langle n+1| + \mathrm{H.c.} \right)$$

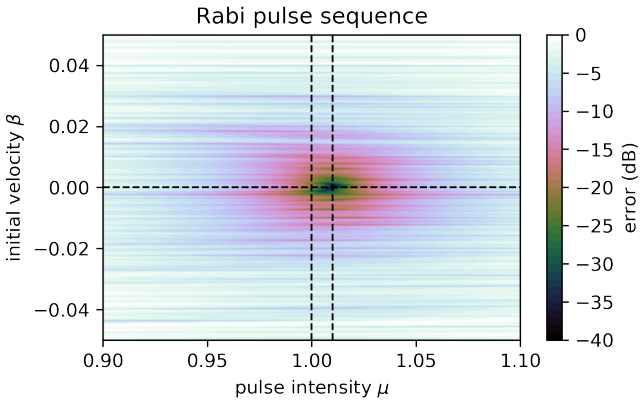
Robustness?

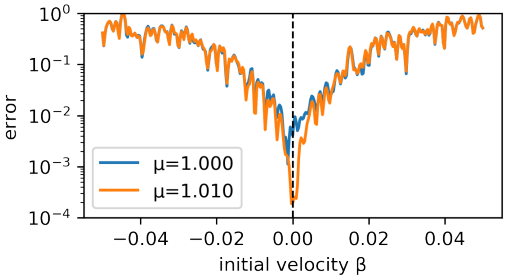
$$\hat{\mathrm{H}} = \sum_n \left(E_n^{(0)}(t) + 2n\beta\hbar\omega_k \right) |n\rangle\langle n| - \mu S(t) \left(|n\rangle\langle n+1| + \mathrm{H.c.} \right)$$

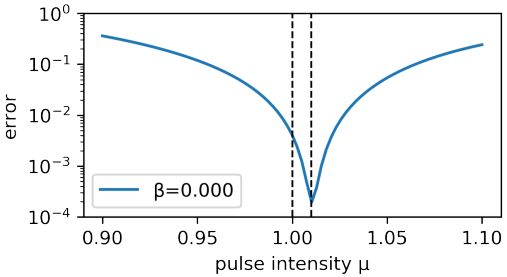
noise parameters:

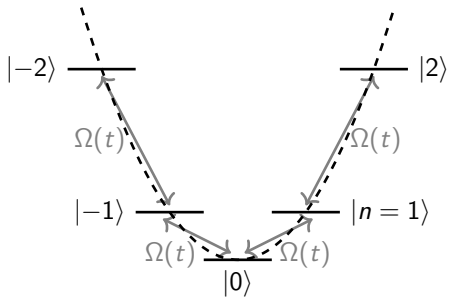
- μ (pulse intensity),
- $\beta = p_0/2\hbar k$ (initial velocity)

Robustness of sequential Rabi pulses









$$E_n(t) \approx n^2\hbar\omega_k + 2n\beta\hbar\omega_k + n\hbar\dot{\phi}(t))$$

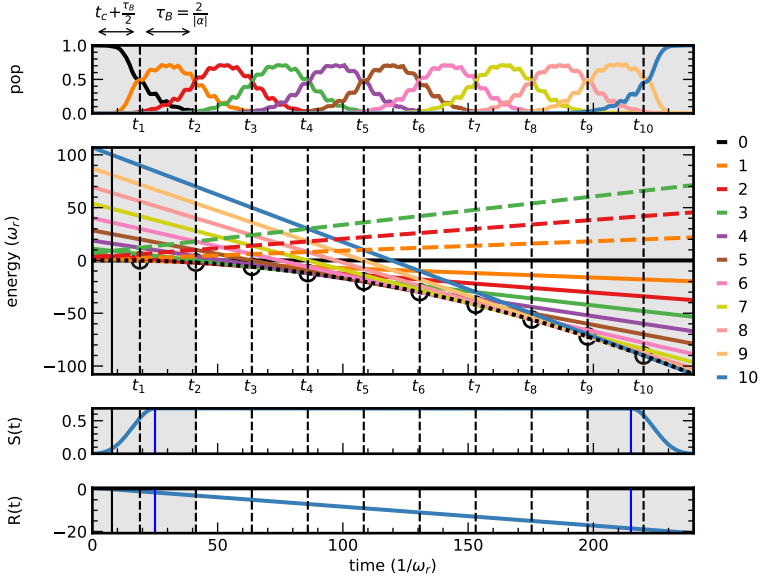
$$\omega_k = 2\hbar k^2/m$$
$$\beta = p_0/2\hbar k$$

Rapid Adiabatic Passage (RAP)

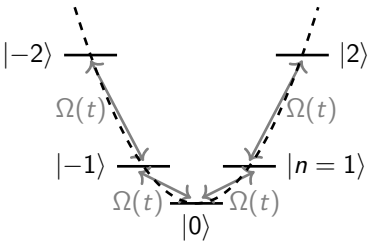
Linear frequency chirp running over transitions

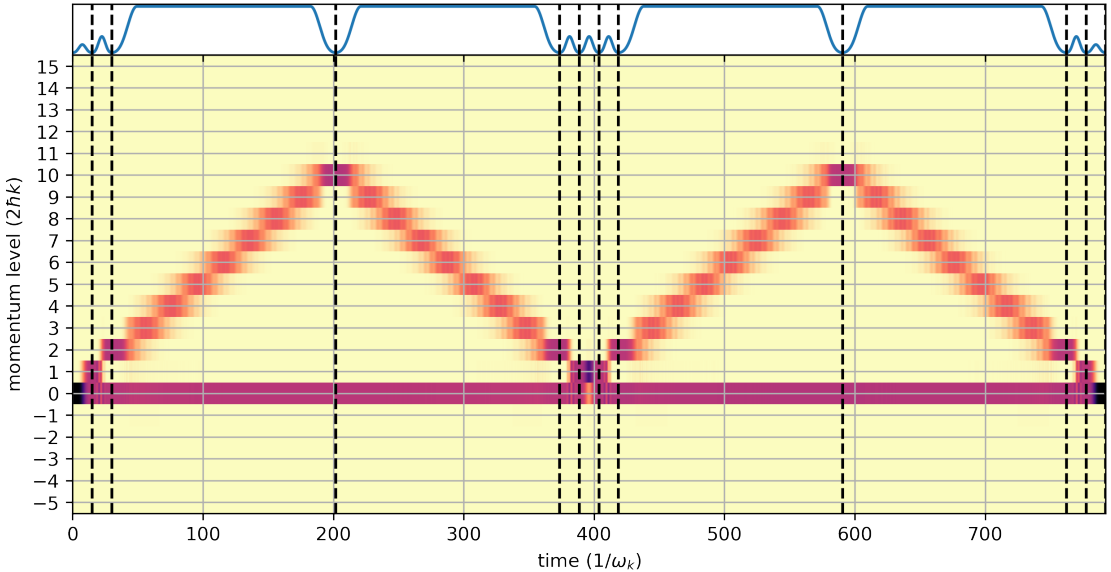
$$\phi(t) = -\frac{1}{2}\alpha\omega_k(t - t_c)^2$$

Malinovsky, Berman. *Phys. Rev. A* **68**, 023610 (2003)

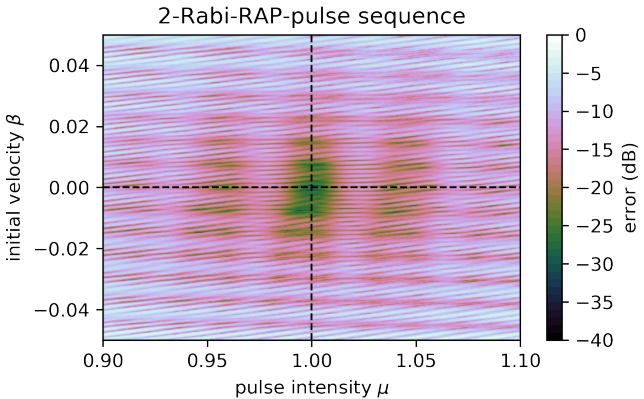


RAP population transfer

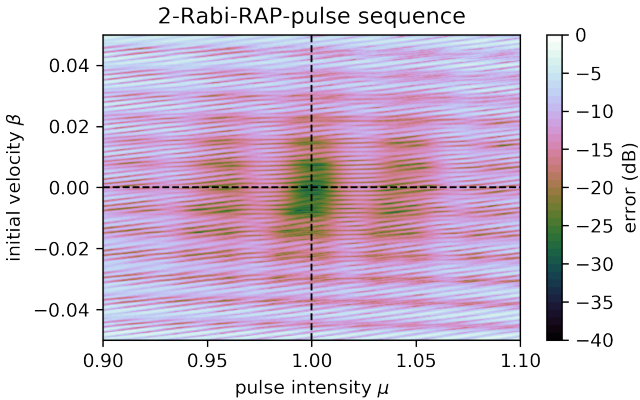


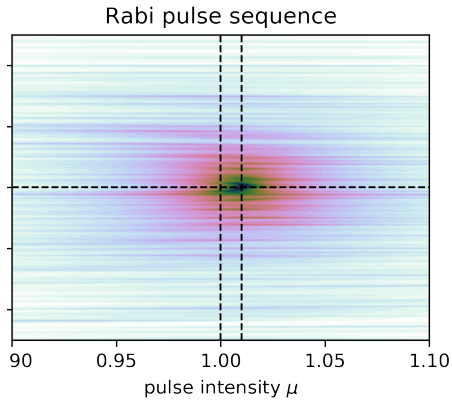

momentum level ($2\hbar k$)
15
14
13
12
11
10
9
8
7
6
5
4
3
2
1
0
−1
−2
−3
−4
−5
0
100
200
300
400
500
600
700
time ($1/\omega_k$)

Robustness of RAP pulse scheme

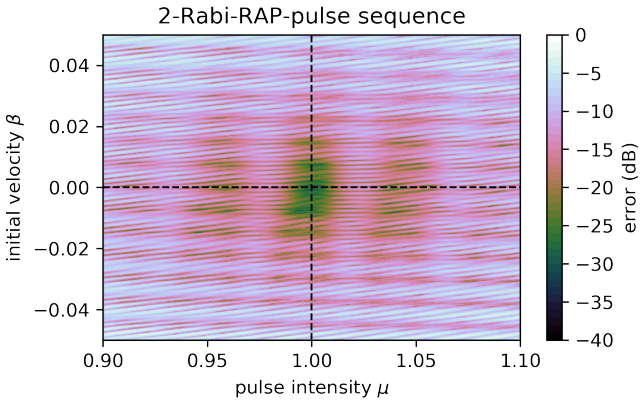


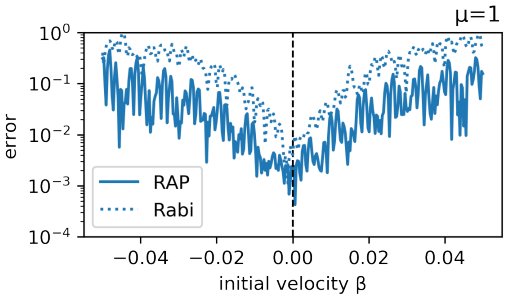
Robustness of RAP pulse scheme

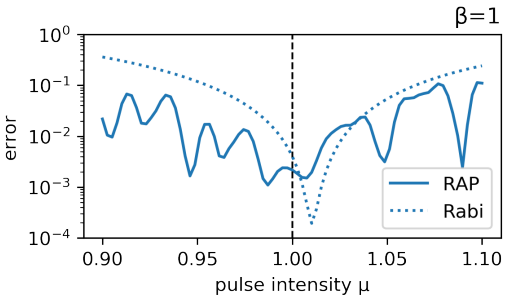




Robustness of RAP pulse scheme







numerical optimal control:

Iteratively minimize figure of merit

$$J_{T,\mathrm{sm}} = \varepsilon_{\mathrm{sm}} = 1 - |\langle \Psi^{\mathrm{tgt}} | \Psi(T) \rangle|^2 ;$$

Find $S(t)$ that ensures $|0\rangle \rightarrow |\Psi^{\mathrm{tgt}}\rangle = |10\rangle$

Implementation of Krotov's method: `github.com/qucontrol/krotov`

Goerz et al., *SciPost Phys.* **7**, 80 (2019)

numerical optimal control:

Iteratively minimize figure of merit

$$J_{T,\mathrm{sm}} = \varepsilon_{\mathrm{sm}} = 1 - |\langle \Psi^{\mathrm{tgt}} | \Psi(T) \rangle|^2 ;$$

Find $S(t)$ that ensures $|0\rangle \rightarrow |\Psi^{\mathrm{tgt}}\rangle = |10\rangle$

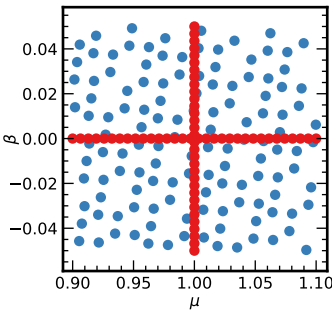
Implementation of Krotov's method: `github.com/qucontrol/krotov`

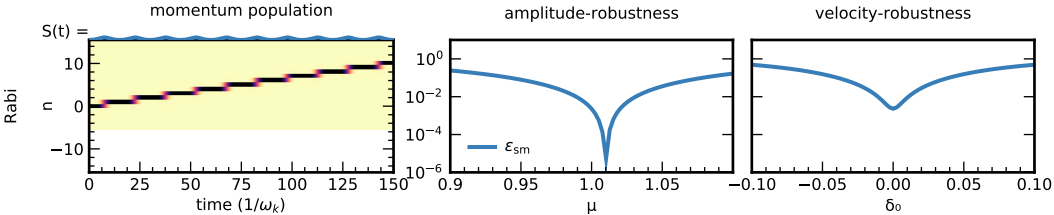
Goerz et al., *SciPost Phys.* **7**, 80 (2019)

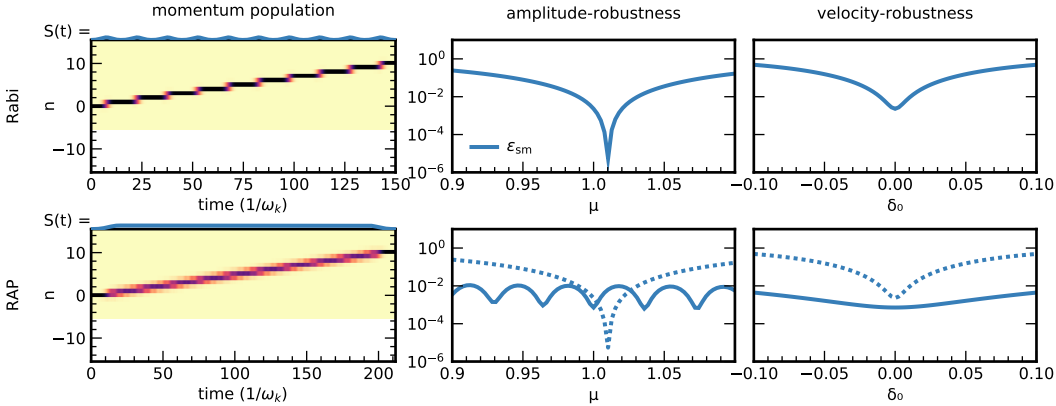
ensemble optimization

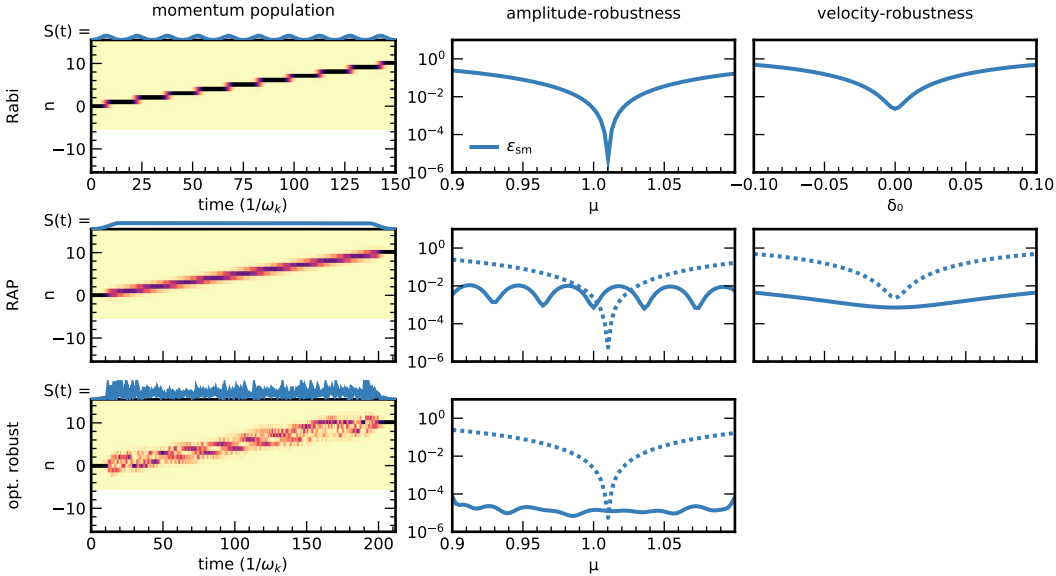
simultaneously optimize over average of random noise realizations

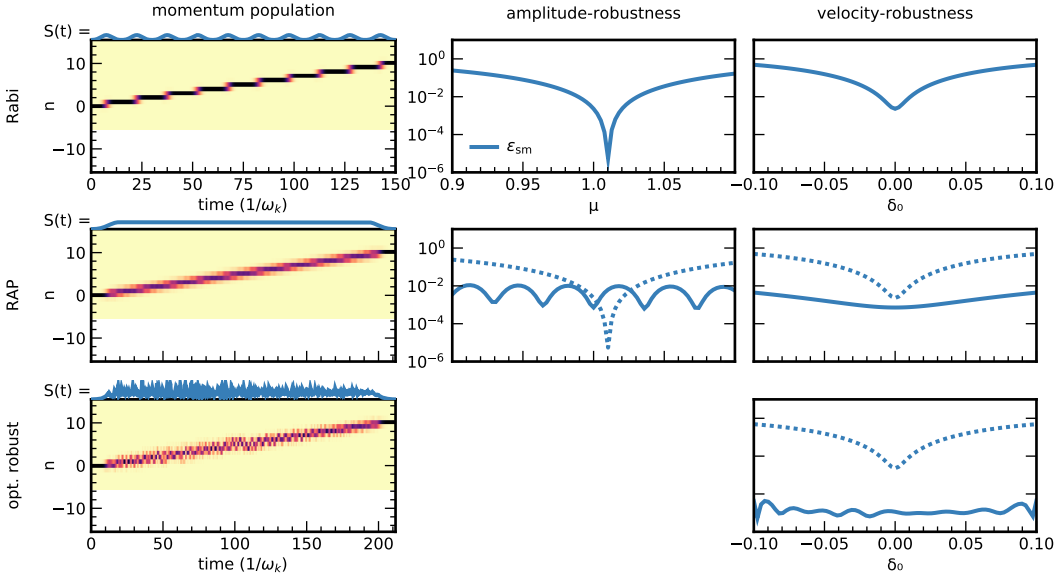
Goerz et al., *Phys. Rev. A* **90**, 032329 (2014)

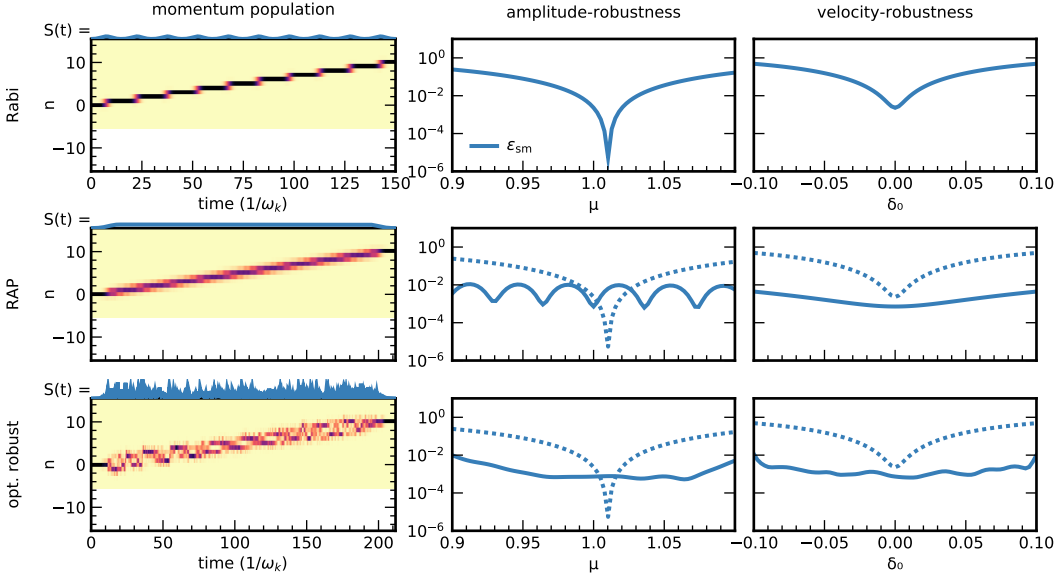



momentum population
S(t) =
Rabi
n
10
0
−10
0
25
50
75
100
125
150
time ($1/\omega_k$)
amplitude-robustness
10^0
10^{-2}
10^{-4}
10^{-6}
ϵ_{sm}
0.9
0.95
1.0
1.05
μ
velocity-robustness
−0.10
−0.05
0.00
0.05
0.10
δ_0


momentum population
amplitude-robustness
velocity-robustness
Rabi
RAP
S(t) =
n
time ($1/\omega_k$)
ϵ_{sm}
μ
δ_0


momentum population
amplitude-robustness
velocity-robustness
Rabi
RAP
opt. robust
S(t) =
n
time ($1/\omega_k$)
ϵ_{sm}
μ
δ_0


momentum population
amplitude-robustness
velocity-robustness
Rabi
RAP
opt. robust
S(t) =
n
time ($1/\omega_k$)
ϵ_{sm}
μ
δ_0


momentum population
amplitude-robustness
velocity-robustness
Rabi
RAP
opt. robust
S(t) =
n
time ($1/\omega_k$)
ϵ_{sm}
μ
δ_0

Summary

- Analyzed robustness for train of Rabi pulses
- More robust: Rapid Adiabatic Passage
- Order of magnitude improvement in robustness: **ensemble optimization**

Outlook

- Optimized scheme for full interferometer
- Experimental constraints

Thank you!